FORTY-FOUR UNIT SQUARES NEED A SQUARE OF SIDE SEVEN

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Abstract. Let $s(n)$ be the side of the smallest square into which n unit squares with pairwise disjoint interiors can be packed, rotations allowed. We prove $s(44) = 7$. The proof is a weighted closed cover of $[0,7]^2$: an explicit measure made of 15,525 weighted points and 3,912 uniformly weighted pieces of the interior grid lines, of total mass $2199755099/50000000 < 44$, such that every closed unit square in $[0,7]^2$ has mass at least one. The covering property is verified by two independently written exact rational-arithmetic checkers, one of them both with and without symmetry reduction. The certificate has only 0.4% of slack, and finding it required a new separation step for low-mass "valleys" formed by nearly axis-parallel squares resting on the walls. We describe that step because the same obstruction should appear in other cases with a thin margin.

1. The result

Write $s(n)$ for the side of the smallest square containing n unit squares with pairwise disjoint interiors, each free to rotate. Since s is nondecreasing and $s(k^2) = k$, every n with $(k-1)^2 < n \le k^2$ has $s(n) \le k$, and a natural question is how many squares can be removed from the $k \times k$ grid before a smaller container suffices. Friedman's survey [5] records the history of the problem. Nagamochi [11] stated $s(k^2-1) = s(k^2-2) = k$, but his Lemma 1 is false [8]; $s(k^2-1) = k$ has been re-proved by Karakuş [8], and $s(k^2-2) = k$ for $k \ge 4$ follows from the results on $s(k^2-3)$ below. The family $s(k^2-3) = k$ is due to Kearney and Shiu for $k = 3$ [9], Bentz for $4 \le k \le 7$ [1, 2], and Daniel for all $k \ge 6$ [4], who also proved $s(k^2-4) = k$ for all $k \ge 5$ by computer-assisted weighted covers (the case $k = 9$ was first proved by Hosono; see the record in [4]). Hosono proved $s(59) = 8$ [7]. Before this note the best bounds for $n = 44$ were $6.9725 \le s(44) \le 7$ [13].

Theorem 1. $s(44) = 7$.

The upper bound is the 7×7 grid with five cells empty. The lower bound follows from two standard lemmas and one finite certificate.

2. The method

For $c \in \mathbb{R}^2$, an angle θ and a side $L > 0$, let $S(c, \theta, L)$ be the closed square of side L centered at c with sides at angle θ to the axes, that is, the set of points whose coordinates (u, v) in the frame rotated by θ about c satisfy $|u|, |v| \le L/2$; its interior is given by strict inequalities. A *pose* in $[0,k]^2$ is a closed unit square $S(c, \theta, 1) \subseteq [0,k]^2$.

The two lemmas below are the standard reduction behind every lower bound proved by unavoidable sets, from Göbel's original argument [6] through Stromquist [12] and the survey [5]; Nagamochi [11] already weighted points, segments and areas, and the use of weights found by linear programming, checked in exact arithmetic, comes from Burns [3] and Massaccesi [10]. We use them in the closed form of Daniel [4], where both are proved in Lean 4 with Mathlib (`packing_le_measure` and `not_packs_of_measure` in `lean/Sqpack/MixedMeasure.lean`). We include the short proofs to keep the note self-contained.

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Lemma 2 (scaling)**.** *If n unit squares with pairwise disjoint interiors lie in a square of side $s < k$, then there are n poses in $[0,k]^2$ that are pairwise disjoint as closed sets.*

Proof. Translate the container to $[0,s]^2$ and apply the homothety $x \mapsto \lambda x$ with $\lambda = k/s > 1$. It maps the container onto $[0,k]^2$ and the i-th unit square $S(c_i,\theta_i,1)$ onto $S(\lambda c_i,\theta_i,\lambda)$, and it maps interiors to interiors, so the n squares $S(\lambda c_i,\theta_i,\lambda)$ still have pairwise disjoint interiors and lie in $[0,k]^2$. Now let $P_i = S(\lambda c_i,\theta_i,1)$ be the concentric unit square with the same orientation. A point of P_i has rotated coordinates with $|u|,|v| \le \frac{1}{2} < \frac{\lambda}{2}$, so P_i lies in the interior of $S(\lambda c_i,\theta_i,\lambda)$. Hence the P_i are pairwise disjoint, and each lies in $[0,k]^2$. □

The closedness matters. Two tiles of the $k \times k$ grid that share an edge both contain that edge, so a measure that counts closed squares may put mass on the grid lines; Lemma 2 is what allows us to assume that no two squares of a packing at side $s < k$ share a point.

Lemma 3 (weighted counting)**.** *Let μ be a finite Borel measure on $[0,k]^2$ with $\mu(P) \ge 1$ for every pose P. Then at most $\mu([0,k]^2)$ poses can be pairwise disjoint.*

Proof. Closed squares are Borel sets. If $P_1,\dots,P_m$ are pairwise disjoint poses, then by additivity $m \le \sum_{i=1}^{m} \mu(P_i) = \mu\big(\bigcup_i P_i\big) \le \mu([0,k]^2)$. □

Corollary 4. *If μ is as in Lemma 3 and $\mu([0,k]^2) < n$, then $s(n) \ge k$.*

Proof. If $s(n) < k$, some square of side $s < k$ holds n unit squares with disjoint interiors. Lemma 2 gives n pairwise disjoint poses in $[0,k]^2$, and Lemma 3 gives $n \le \mu([0,k]^2) < n$. □

All our measures are finite sums of point masses and of measures spread uniformly by length on axis-parallel segments; such a measure is *valid* on $[0,k]^2$ if every pose has mass at least one.

Remark 5 (a limit of the method). By linear-programming duality, the least total mass of a valid measure is at least the fractional packing number of $[0,k]^2$. A cover can therefore prove $s(n) = k$ only when that number is below n. For $n = 12$ and $k = 4$ it is at least 12.2688 [4], so no cover of this kind proves $s(12) = 4$. For $n = 44$ and $k = 7$ our search puts the optimum at about 43.82, which leaves 0.4% of room.

3. The certificate

The file `s44_cover_7.txt` [14] (SHA-256 `c9fde0f1...ba254cae0`), written in Daniel's mixed-cover format [4], lists, with coordinate denominator 1000 and mass denominator 10^8,

- 15,525 point masses, of total 11.714655, and
- 3,912 pieces of the lines $x = i$ and $y = i$, $1 \le i \le 6$, each carrying its mass uniformly by length, of total 32.280447.

The total is exactly $2199755099/50000000 = 43.99510198 < 44$. The measure is invariant under the eight symmetries of the square, as checked exactly by the script `sanity.py` in [14]. The claim to verify is:

($\star$) *every closed unit square $P \subseteq [0,7]^2$, at every center and angle, has $\mu(P) \ge 1$.*

Given ($\star$), Corollary 4 with $n = 44$ and $k = 7$ gives $s(44) \ge 7$, which proves Theorem 1.

4. Verification

The space of poses is three-dimensional (center and angle). Both checkers subdivide it into boxes and certify each box in exact rational arithmetic, using primitives that have paper proofs in [4]: a point, or a chord of a segment, that lies in every square of the box; lemmas for squares against the walls; and a threshold lemma for squares close to a grid tile, where the mass of a measure on lines jumps as the angle tends to 0.

checker	mode	result	time
`zmx2` (Rust)	D4-reduced	VERIFIED, 0 uncertified	191 s
`zmx2` (Rust)	full, no symmetry	VERIFIED, 0 uncertified	27.6 min
`zm_mixed.py` (Python)	D4-reduced, depth 24, then 30	VERIFIED, 0 uncertified*	≈143 CPU-h

*All 9,800 root boxes were swept at depth 24. That sweep left 1,265 boxes in 27 roots uncertified, and each of those 27 roots was swept again in full at depth 30, with no uncertified boxes (deepest leaf at depth 29). Floating-point minima of $\mu(P)$ in those roots range from 1.0008 to 1.0041.

Both exact checkers are by Daniel [4]. They were written separately and were used unmodified at commit `659ee62`. They share the closed-square conventions and the reduction by the eight symmetries; the full `zmx2` sweep does not use that reduction. As a check that shares no code with either, a floating-point evaluation of $\mu(P)$ at 675,000 random, near-tile and near-45° poses found a minimum mass of 1.0031. All logs are in [14].

5. How the certificate was found

We solved the linear program "minimize $\mu([0,7]^2)$ subject to $\mu(P) \geq 1$" over symmetric point and segment variables by constraint generation. Each round solves the LP, scales the solution to total 43.995, runs `zmx2`, and adds the poses of uncertified boxes as new constraints. The starting point was Daniel's cover proving $s(45) = 7$ [4].

Near-wall valleys. Progress stalled after four rounds. The worst floating-point coverage in uncertified boxes crept from 0.975 to 0.992, and every failure sat at the wall tile centered at $(3/2, 1/2)$ and its images. The cause is a family of poses that the standard separation misses. A nearly axis-parallel square resting on a wall beside a corner tile, tilted by $\theta \in [0.01°, 3°]$ and displaced by about $0.28\,\theta$ along the wall, captured only 0.989. The lattice separation (pitch 0.01, steps of 0.5°) steps over such valleys, and the family of near-tile poses samples only $\theta = 10^{-5}$. With a slack of several percent, scaling the cover up hides them; with 0.4% it does not.

Our separation step, `tilesep.py`, scans for each tile in a fundamental domain both signs of the tilt, 36 tilts in $[10^{-5}, 0.05]$, and displacements $\theta \cdot (t, u)$ on a grid, keeping only poses inside the container, and then refines the worst ones locally. Once it was added, the number of uncertified roots fell $141 \to 88 \to 46 \to \cdots \to 7 \to 0$ within ten rounds (with one setback, to 181, after a restart that dropped temporary constraints), while the LP value rose only from 43.806 to 43.824.

6. Remarks

Even the optimal cover is only 0.4% below 44, and our certificate's total is within 0.011% of 44. Searches at a comparable margin should expect obstructions, like the valleys above, that uniform sampling does not see; here, separation aimed at the geometry of the walls and the grid tiles made the difference.

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